

Sophia Zhang

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

B.S. in Computer Science and Mathematics

Aug. 2025 – May 2028 (expected)

- Coursework: Inference & Information (G), Probability (G), Machine Learning (G), TinyML and Efficient AI Computing (G), Operating Systems, Computation Structures

Princeton International School of Mathematics and Science (PRISMS)

Princeton, NJ

High school

Sep. 2022 – June 2025

EXPERIENCE

Undergraduate Researcher (UROP)

Sep. 2025 – Present

Dina Katabi's Lab, MIT CSAIL

Cambridge, MA

- Working with diffusion models, multimodal foundation models, and recommendation models.

AI Engineering Intern

June – Aug. 2026

Alpha School

Austin, TX

- Built RL (GRPO) training setups and evaluations for LLM tutors and educational apps, grounded in learning science and pedagogy.

Trading & Technology Intern

Jan. 2026

Hudson River Trading

New York, NY

- Software engineering and quant research projects, including trading Brazilian equities.

Research Collaborator – LLMs for Formal Theorem Proving

June 2025 – Jan. 2026

Microsoft Research Asia

Remote

- Verified Lean formalizations of math proofs and contributed to system improvements; two papers below.

PUBLICATIONS

STAR-PólyaMath: Multi-Agent Reasoning under Persistent Meta-Strategic Supervision

NeurIPS 2026 (Spotlight)

- Jiaao Wu, Xian Zhang, Hanzhang Liu, **Sophia Zhang**, Fan Yang, Yinpeng Dong

Euclean: Automated Geometry Problem Formalization with Unified Verification in Lean

ICML 2026

- Linbin Tang, Jingyan You, Zilin Kang, Hanzhang Liu, **Sophia Zhang**, Zenan Li, Chenrui Cao, Liangcheng Song, Jiaao Wu, Xian Zhang, Fan Yang

PROJECTS

AI Math Olympiad Progress Prize 3 | *Python, PyTorch, vLLM, Unsloth*

Jan. – Apr. 2026

- Competed with a team of MIT freshmen, all Olympiad medalists. Built for the training and inference scaffolding (vLLM, Unsloth), hidden-state extraction, and parameter sweeps.
- Authored the team's technical writeup.

AWARDS

Putnam Top 200, 2026

USAMO Silver, 2025

USAPhO Honorable Mention, 2025

Mathematical Olympiad Program (MOP) Two-time attendee, 2023, 2024

TECHNICAL SKILLS

Languages: Python, C++, C, Rust, Java, TypeScript, Lean, RISC-V Assembly, Minispec, Dafny

AI Research: RL post-training, diffusion models, multimodal models, LLM agents & multi-agent reasoning, formal theorem proving, recommendation models

Tools: PyTorch, vLLM, Unsloth, NumPy, SciPy, Pandas, Matplotlib, Jupyter, automated eval pipelines, AI coding agents (Claude Code) & multi-agent workflows